

Modeling Pedestrian Walking Behavior Parameters and Pedestrian Safety at Congested Urban Intersections in Educational and Recreational Zones for SDGs-2030

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Abstract: In developing countries like India, especially in well planned cities such as Chandigarh, pedestrian safety at urban intersections is shaped by factors including population density, land-use patterns, public transport availability, and surrounding activity intensity. Improving pedestrian safety at congested intersections requires integrating these context-specific factors into traffic planning, design, and management. This study aims to modeling pedestrian Walking behaviour parameters and enhance pedestrian safety through a comprehensive, evidence-based approach aligned with Sustainable Development Goal (SDG) 11, which emphasizes inclusive, safe, resilient, and sustainable urban development. The study employs a robust methodology involving detailed field observations and systematic data collection at selected high-congestion intersections. Key parameters influencing pedestrian behavior such as traffic flow, pedestrian volume and density, signal control, and crossing characteristics are analyzed to understand movement patterns and interaction dynamics. Both quantitative and qualitative analyses are used to capture behavioral complexity in diverse urban settings. The findings offer valuable insights for urban planners and policymakers, supporting informed decisions for developing safer, more efficient, and pedestrian-friendly infrastructure. The study highlights the importance of context-sensitive, location-specific interventions to improve pedestrian safety, traffic operations, and sustainable urban mobility for all road users.

Keywords: Pedestrian, effective walkway, urban planning, flow rate, walkway width

1. General

1.1 Scope of Study

Traffic accidents involving pedestrians have become a major global safety concern, especially in developing countries with high population density, rapid urbanization, and non-compliance with traffic rules by both drivers and pedestrians. Pedestrian crossings, which are meant to provide safety, have become a common site of accidents, either when drivers neglect traffic rules or when pedestrians try to force their way through the traffic stream. Additionally, high foot traffic at pedestrian crossings can cause unacceptable delays for vehicles. Pedestrians are a significant contributor to urban traffic accidents, and the need for targeted interventions and urban planning strategies to enhance pedestrian safety is urgent [1].

Our research on pedestrian behaviour is critical in developing holistic strategies that address the multifaceted challenges associated with pedestrian safety in India's diverse urban landscape. By merging the insights from the pedestrian accident study, we can contribute to making pedestrian crossings safer and reducing the number of accidents in India and other developing countries to achieve Sustainable Development Goal 11, 2030.

1.2 Traffic in India

The release of a comprehensive study on pedestrian accidents in India during the 7th UN Global Road Safety Week sheds light on this critical issue. The study, which analysed 6,300 accidents through the Road Accident Sampling System for India (RASSI), revealed that pedestrian accidents account for nearly 10% of traffic fatalities in the country. This alarming statistic underscores the need for targeted interventions and urban planning strategies to enhance pedestrian safety in diverse urban settings.

The study also found that rural roads pose a heightened risk to pedestrians, accounting for 56% of all pedestrian accidents. This finding reinforces the importance of our research, which explores pedestrian dynamics in various zones and aligns with the report's identification of challenges in rural, urban, and semi-urban environments. Contrary to common belief, the study challenges the notion that most accidents occur at night, with 57% of pedestrian accidents happening during the day. This resonates with our exploration of pedestrian behaviour throughout different times, providing a more nuanced understanding of risk factors.

The study also highlights that human error is the primary cause of pedestrian accidents, emphasizing the significance of behavioural factors in shaping pedestrian movement. Additionally, the study's unique observation of pedestrians stopping in the middle of the road to let vehicles pass adds a valuable cultural dimension. This underscores the need for culturally sensitive interventions in urban planning that consider local practices and habits.

1.3 Traffic in Chandigarh

As a city, Chandigarh has emerged as a remarkable success story by providing its inhabitants with a high standard of living, accompanied by substantial growth. This has made it an attractive destination for not only middle and upper-class families but also skilled and unskilled workers. According to estimates, the city's urban population in 2011 was 10.55 lakh, with the villages surrounding the city growing even faster. With the rapid growth of satellite towns, such as Panchkula and Mohali, coupled with the increase in living standards.

Table 1. Data on Pedestrian and Cyclist Accidents in Chandigarh[6]

Road Crashes Data -2018-2023														
Year	Total Cases	Fatal Cases	Killed Persons		Pedestrian Deaths		Cyclist Deaths		Two-Wheeler Deaths		Non-Fatal Cases	Injured Persons		U/s 174
														CrPC
2018	316	97	98		35		11		40		219	300		21
			M	F	M	F	M	F	M	F		M	F	
			83	15	27	8	10	1	35	5		232	68	
2019	305	100	104		36		10		46		205	275		29
			M	F	M	F	M	F	M	F				
			84	20	25	11	8	2	41	5		216	59	
2020	159	50	53		12		10		20		109	148		9
			M	F	M	F	M	F	M	F		M	F	
			46	7	9	3	10	0	20	0		117	31	
2021	208	94	96		24		17		32		114	172		14
			M	F	M	F	M	F	M	F		M	F	
			79	17	18	6	15	2	29	3		135	37	
2022	237	79	83		23		12		40		158	203		21
			M	F	M	F	M	F	M	F		M	F	
			68	15	16	7	12	0	35	5		164	39	
2023 upto 27.05.2023	68	24	27		8		6		9		44	67		8
			M	F	M	F	M	F	M	F		M	F	
			21	6	5	3	5	1	7	2		53	14	

***CrPC: Code of Criminal Procedure is the main legislation on procedure for the administration of substantive criminal law in India.**

There has been an unprecedented surge in motor vehicle ownership in recent years, resulting in a substantial increase in the number of vehicular trips made. The number of cars on the city's roads is increasing daily, leading to a steep rise in traffic. Nonetheless, the city's street system has some glaring defects, including insufficient junction spacing and arbitrary treatment at significant intersections. As per available data, as many as 19.71% of fatal accidents in Chandigarh involve cyclists. Additionally, the variation of speed for different categories of traffic exacerbates the severity of accidents and the number of conflicts with pedestrians.

1.4 Design for Pedestrian Facilities

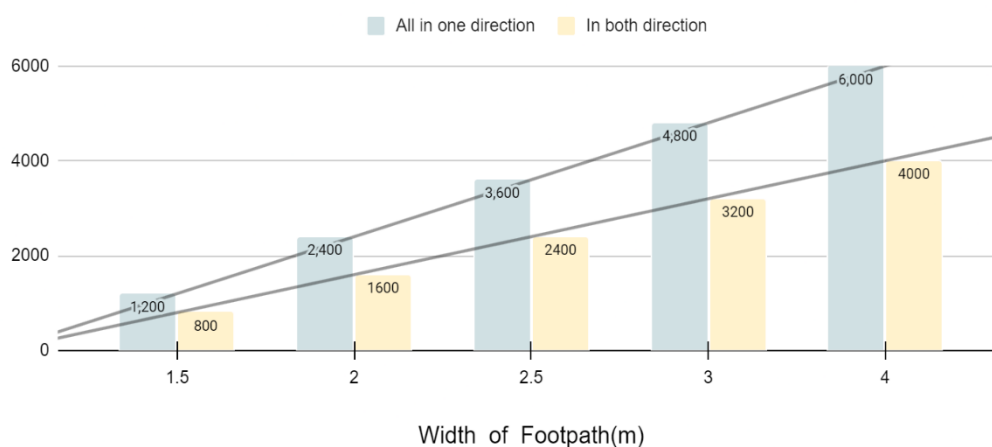
To ensure the safety of pedestrians, it is essential to provide separate footpaths next to the carriageway on urban streets. These footpaths should be located on both sides of the road, and ideally, they should be raised above the street level. To encourage people to use the footpaths, they should be well-paved and maintained properly. They should also be sloped appropriately to allow rainwater to drain away. A crossfall within the range of 1 in 40 is considered sufficient. For the safety of pedestrians, separate footways should be provided alongside the carriageway of urban streets. They should be provided on either side of the road and should preferably be raised above the general carriageway level. To attract full-use pedestrians, they should be properly surfaced or paved. They should be sloped adequately to drain away the rainwater. Crossfalls within the range of 1 in 40 are satisfactory.

Table 2. Recommended Minimum Footway Width as per U.K. standards. [2]

Type of road	Recommended minimum footway width
Primary Distributor	
Urban Motorway	No footway
All-purpose road	2.7m
District distributor	2.7 m in principal business and industrial districts
Local distributor	2.4 in residential districts.
	2.7 m in principal business and industrial districts
Access road	1.8 m in residential districts
	The principal mean of access: is 2.7 m in principal business districts
	1.8 m in industrial districts
	1.8 m normally in residential districts
	3.7-4.5 m adjoining shopping frontage.
	The secondary mean of access: 0.9 m verge instead of footway on road in principal business and industrial districts.
	0.6 m verge instead of footway on road in residential districts

Table 3. Recommended Minimum Footway Width as per I.R.C guidelines. [3]

Width of Footpath (m)	Capacity in number of people per hour	
	All in one direction	In both direction
1.5	1,200	800
2	2,400	1,600
2.5	3,600	2,400
3	4,800	3,200
4	6,000	4,000

Capacity in number of person per hour

1.5 Highway Capacity Manual Methodology [4]

The proposed methodology presents a comprehensive framework for the critical evaluation of pedestrian facilities. Using this framework, analysts are empowered to investigate the impact of bicycles and traffic signals on pedestrian facilities and the effect of pedestrian volume on flow and level of service (LOS).

1.6 Pedestrian LOS

HCM analyses pedestrian LOS based on pedestrian flow rate and sidewalk space. As density increases, pedestrian speed decreases, and mobility declines. Pedestrian LOS is calculated using pedestrians per minute per meter (ped/min/m). Effective sidewalk width is calculated by subtracting obstacle widths and a 0.30 to 0.45 m buffer width per obstacle from the total width. Using the pedestrian unit flow rate, pedestrian LOS can be calculated using the “Average Flow LOS Criteria for Walkways and Sidewalks.” The platoon LOS table can be used to obtain the platoon LOS. HCM states that platoon LOS is generally one level poorer than that determined by average flow criteria.

1.7 Uninterrupted-Flow Pedestrian Facilities

Pedestrian facilities that are continuous and uninterrupted include both separate and shared pedestrian paths, both indoors and outdoors, that are designated for pedestrian use. These pedestrian facilities are unique because pedestrians do not experience any disruption, except for interaction with other pedestrians, and, on shared paths, with other non-motorized modes of transportation. These procedures should consider pedestrian walking speed, pedestrian start-up time, and pedestrian space requirements to ensure efficient pedestrian flow.

1.8 Walkways and Sidewalks

Pedestrian walkways and sidewalks are designed to keep pedestrians safe by separating them from motor vehicle traffic. Bicycles or any other type of user, besides pedestrians, are not allowed on these paths. Such facilities are commonly built on city streets, airports, in subways, and bus terminals. These pedestrian facilities include straight sections of sidewalk, terminals, stairs, and crosswalks where streams of pedestrians cross.

Table 4. Criteria for determining the average flow LOS of walkways and sidewalks.

LOS	SPACE(M/p)	FLOW RATE(P/MIN/M)	SPEED(M/S)	V/C RATIO
A	>5.6	<=16	>1.30	<=0.21
B	>3.7-5.6	>16-23	>1.27-1.30	>0.21-0.31
C	>2.2-3.7	>23-33	>1.22-1.27	>0.31-0.44
D	>1.4-2.2	>33-49	>1.14-1.22	>0.44-0.65
E	>0.75-1.4	>49-75	>0.75-1.14	>0.65-1.0
F	<=0.75	Variable	<=0.75	Variable

2.0 Determining Pedestrian Walking Unit Flow Rate

To make field observations simpler, the pedestrian unit flow rate is used to measure service. To calculate the pedestrian unit flow rate, the peak 15-minute count and the effective walkway width must be determined. The formula for calculating pedestrian unit flow rate is $v_p = v_{15} \times W_e \times 15$, where v_p is the pedestrian unit flow rate (p/min/m), v_{15} is the peak minute flow rate (p/15-min), and W_e are the effective walkway width (m).

$$v_p = v_{15} / 15 * W_e$$

where v_p = pedestrian unit flow rate (p/min/m),

v_{15} = peak 15-min flow rate (p/15-min), and W_e = effective walkway width (m).

2.1 Determining Pedestrian Walking Space

When it comes to evaluating walkways and sidewalks, the most important factor to consider is space, which is the opposite of density. Space can be measured on-site by calculating the

facility's sample area and determining the number of pedestrians that can occupy that area at any given time.

$$\text{Space} = 1 / \text{Density}$$

2.3 Determining Pedestrian Walking Speed

The speed at which pedestrians walk can be affected by the percentage of elderly pedestrians (aged 65 or above) in the walking population. For walkway computations, a walking speed of 1.2 m/s is recommended if 0 to 20 per cent of pedestrians are elderly. However, if the proportion of elderly pedestrians exceeds 20 per cent, a walking speed of 1.0 m/s is recommended. Furthermore, any increase of 10 per cent or more in the number of elderly pedestrians can cause a decrease in walking speed of 0.1 m/s.

2.4 Determining Effective Walkway Width

Calculating the effective walkway width is important as it refers to the portion of the walkway that is usable by pedestrians. Various obstructions can make pedestrians avoid certain areas of the walkway. These obstructions can be found in Figure 1. The effective walkway width can be calculated using the equation below:

$$W_E = W_T - W_O$$

Where W_E = effective walkway width (m),

W_T = total walkway width (m),

W_O = sum of widths and shy distances from obstructions on the walkway (m).

Figure 1 provides a schematic showing common obstacles and the estimated width of the walkway. The values in Figure 1 can be used in cases where specific walkway configurations are not available.

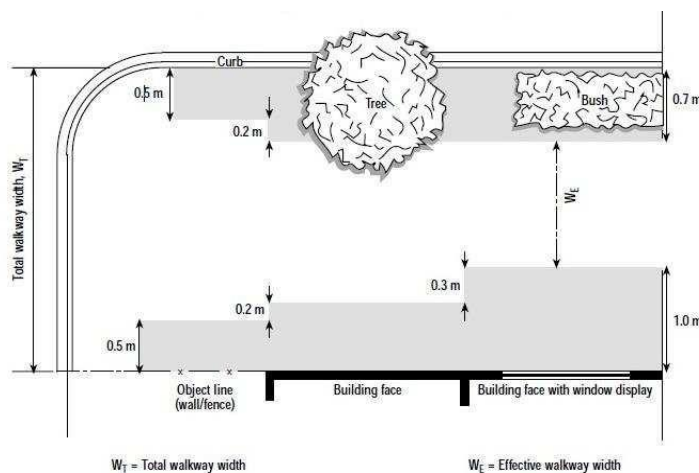


Fig 1 Determination of effective walking width. [5]

To evaluate the effect of occasional obstructions, such as trees or poles, it is customary to assume that their effective length is five times their effective width. To determine the average impact of these obstructions, one can multiply their effective width by the ratio of their effective length to the average distance between them.

Furthermore, it is imperative to consider the presence of right-turning vehicles while pedestrians cross a signalized intersection. These vehicles may occupy a portion of the crosswalk during the crossing phase, rendering it unusable for pedestrians. In such cases, the effective crosswalk width can be determined by subtracting the appropriate time-space used by the right-turning vehicles.

3.0 Madhya Marg Detail Study for Chandigarh City

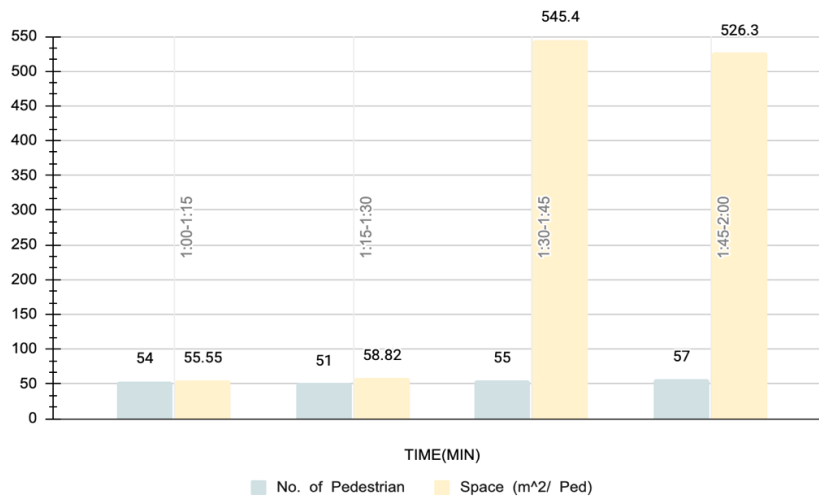
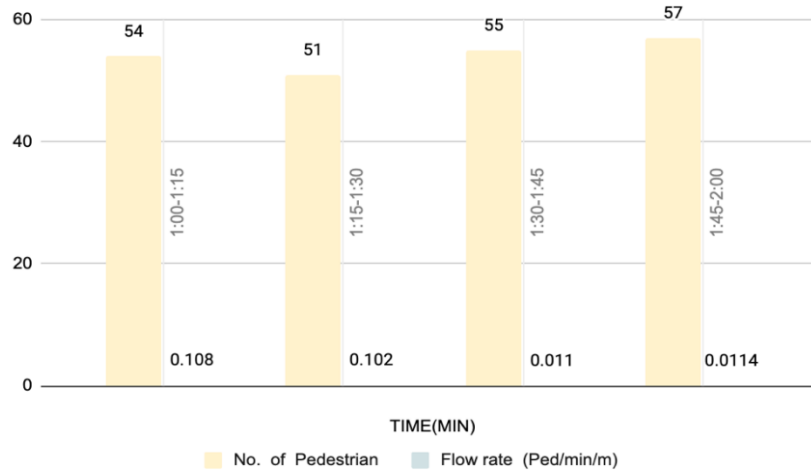
3.1 Recreational Zones

The data was collected on the intersection present near PGIMER, Chandigarh connecting Sector – 11, 12, 14, and 15 during the peak hour. The number of cyclists and other NMT modes has been counted separately on the cycle tracks and the main road. The footpath in front of Nehru Hospital (NH) Block PGIMER is covered with food and accessories stalls. The garbage generated is not maintained well. The conditions accurately depict the location and its current condition.



Table 5. Data of pedestrian count and LOS

Time interval	No. of Pedestrian	Flow rate (Ped/min/m)	Space (m ² /Ped)	Level of service
1:00-1:15	54	0.108	55.55	A
1:15-1:30	51	0.102	58.82	A
1:30-1:45	55	0.011	545.4	A
1:45-2:00	57	0.0114	526.3	A



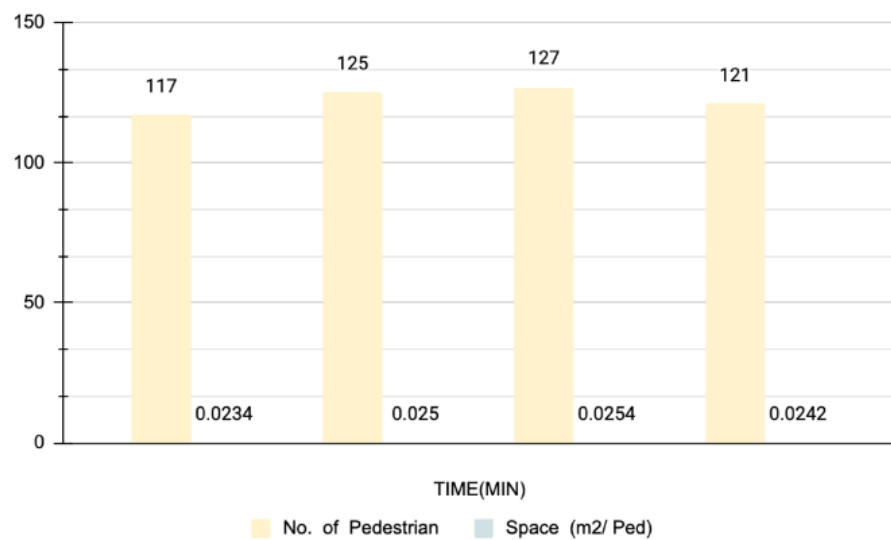
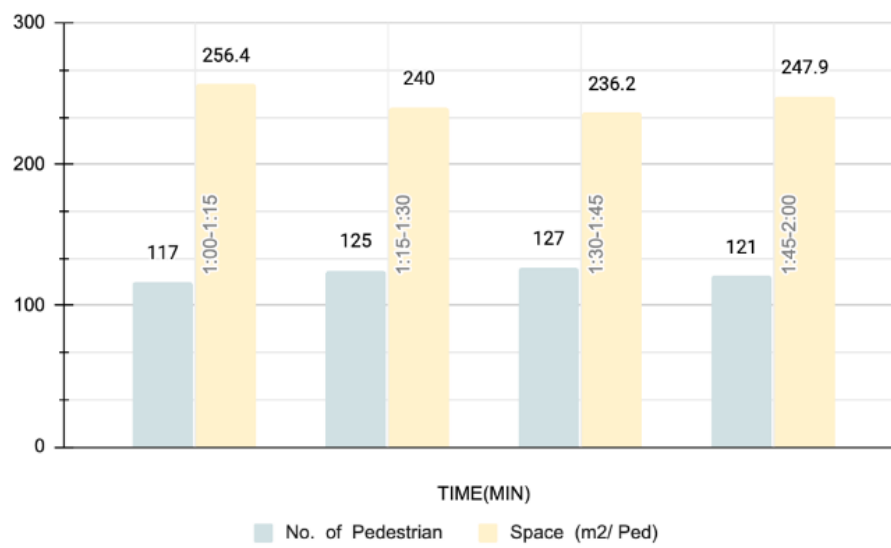
3.2 Educational Zone [5]

The intersection is located in front of Punjab Engineering College on Vidya Path, near the bus stop. The number of cyclists and other NMTs has been counted separately on the intersection, cycle track, and main road of Vidya Path. The footpath and cycle tracks are not well maintained. The dimensions of the 100m section and the photograph accurately represent the location and its current condition.



Table 6. Data of pedestrian count and LOS

Time interval	No. of Pedestrian	Flow rate (Ped/min/m)	Space (m2/ Ped)	Level of service
1:00-1:15	117	0.0234	256.4	A
1:15-1:30	125	0.025	240	A
1:30-1:45	127	0.0254	236.2	A
1:45-2:00	121	0.0242	247.9	A



4. Conclusion

The study conducted on pedestrian and slow traffic behaviour at intersections in densely populated areas within educational and recreational districts is a comprehensive report that reveals the importance of multiple factors that must be considered to ensure the safety and effectiveness of these areas.

The study was conducted in India, with a particular emphasis on Chandigarh, to understand the challenges and complexities of managing pedestrian movement in urban environments across the country. Through this study, several crucial points have been identified that require attention to improve the safety and well-being of pedestrians.

Firstly, the study highlights the alarming statistics related to pedestrian accidents, particularly in densely populated urban areas such as India. Immediate attention and focused responses are needed to improve pedestrian safety. The study suggests that the number of pedestrian accidents can be reduced by implementing measures such as speed limits, pedestrian crossings, and traffic calming devices.

Secondly, the study emphasizes the importance of behavioural factors in ensuring safety and highlights the cultural dimensions of pedestrian interactions with traffic that are crucial for effective urban planning. The study suggests that a better understanding of the cultural nuances of the local population is required to create an effective urban planning strategy that prioritizes pedestrian safety.

Thirdly, the study identifies context-specific interventions as essential and noteworthy. Recognition and acknowledgement of the significance of local practices and habits are required to create a safer and more habitable urban environment. The study suggests that local authorities should work with the community to identify potential hazards and implement strategies that are tailored to the needs of the local population.

Fourthly, the study emphasizes the importance of better traffic management, particularly at busy intersections. Collected and analyzed data highlights the need to assess traffic flow, pedestrian density, and signalization for optimized traffic flows and increased safety for pedestrians. The study suggests that a better understanding of traffic patterns and pedestrian behaviour is required to create an effective traffic management strategy that prioritizes pedestrian safety.

This study highlights the need for a comprehensive and context-dependent approach to improving pedestrian safety in densely populated areas within educational and recreational settings. It involves the establishment of infrastructure, enhancements in traffic management, and behavioural changes that prioritize pedestrian safety and well-being. Ultimately, this approach has the potential to create safer, more efficient, and more habitable urban environments for all residents and visitors.

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